Implementation of Conservation Voltage Reduction (CVR) with PV Farm Inverters as a Strategy to Lower Peak Demand

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1. Final Analysis/Calculations

1.1 Final Network Design

Load flow studies were conducted on two networks. The first network is a realistic distribution (Dx) feeder based on a lab from ECE 4464 (Power Systems II). The second network is the IEEE 14-bus system, which is the smallest standardized test network that represents a more complex power system. Per standard ANSI C84.1 [1], the acceptable bus voltage in a power system is between 0.95 and 1.05 per unit. We selected 0.97 per unit as the target Conservation Voltage Reduction (CVR) voltage for our studies as it was close to the lower band but not the absolute minimum.

1.1.1 Distribution Feeder Design

Figures 1 and 2 show the original and modified Dx feeder networks, respectively. The step-down transformer was moved from between Buses 1 and 2 to between Buses 2 and 3 to ensure the load bus voltage (bus 5) would originally be greater than target voltage of 0.97 per unit. PSSE studies and Python automation were conducted to confirm that the load bus in the modified Dx feeder would remain above 0.97 pu for a peak load of 10MW and various power factors [link]. 10MW was selected as the peak load value to represent a realistic load size for a distribution feeder.

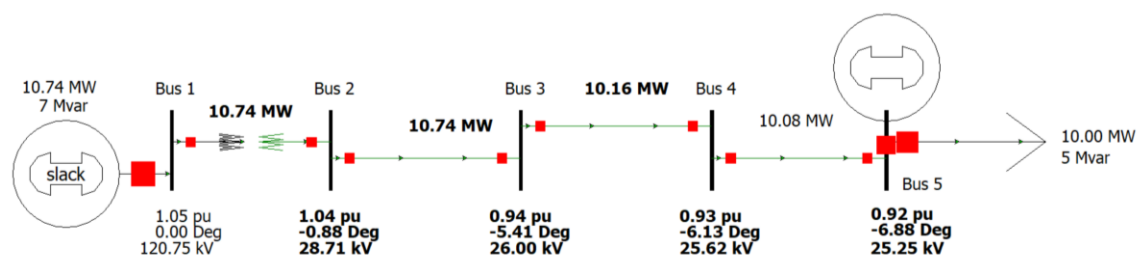

Figure 1: Original Dx Feeder Network from ECE 4464 Lab

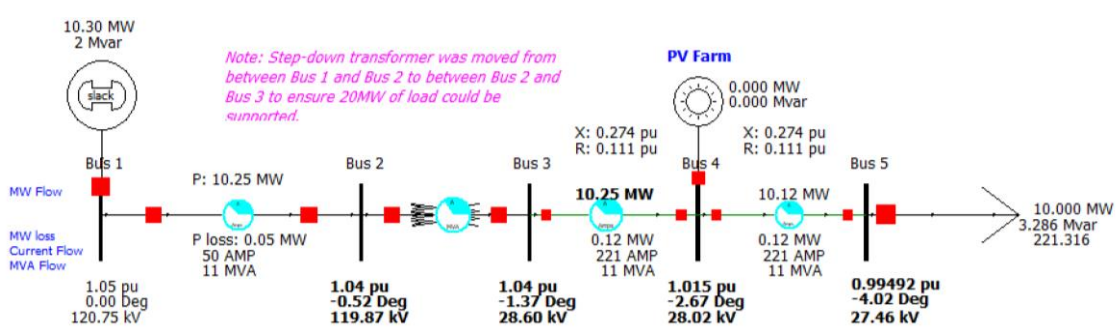

Figure 2: Modified Dx Feeder Network Used in Studies

1.1.2 IEEE 14-Bus System Design

Figure 3 shows the IEEE 14-bus system our group ran studies on. The system data was taken from [2] and slight modifications were made. All system bus voltages above 1.05 per unit were reduced to 1.05 per unit to satisfy standard ANSI C84.1 [1]. Additionally, each load was classified to either “residential”, “industrial”, or “commercial” to model various load behaviour based on [3] (Table 1). The loads at buses 4, 9 and 14 were selected as the focus of our studies (equivalent to Bus 5 in the Dx feeder network) as they were the three largest non-industrial loads. CVR has minimal effect on industrial loads. The load sizes provided in the original data were assumed as the peak values.

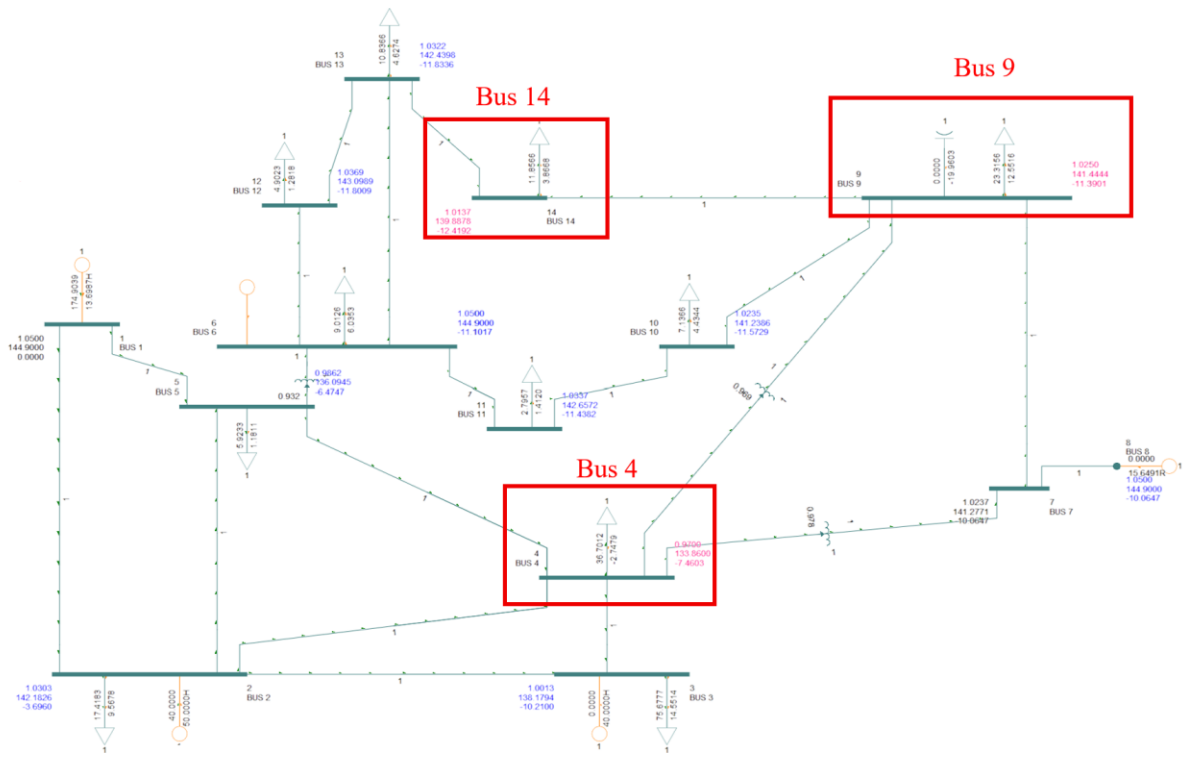

Figure 3: IEEE 14-Bus System in PSSE with Key Load Buses Highlighted

Table 1: Classification and Description of Loads in IEEE 14-Bus System

Bus	Load Type	Load Behaviour
2	industrial	primarily constant power*
3	industrial	primarily constant power*
4	commercial	split between constant power, constant current and constant impedance**
5	commercial	split between constant power, constant current and constant impedance**
6	residential	primarly constant current***
9	residential	primarly constant current***
10	residential	primarly constant current***
11	residential	primarly constant current***
12	residential	primarly constant current***
13	residential	primarly constant current***
14	residential	primarly constant current***

*Constant power loads stay constant regardless of behaviour
**Constant current loads vary linearly with voltage
***Constant impedance loads vary with voltage squared

1.2 Load Curve Calculations

Hourly demand data from the IESO website [4] was analyzed to compute the average load at each hour as a percentage of the peak value (Figure 4). Studies run on the Dx Feeder and IEEE 14-bus system used these percentages to scale the loads appropriately for different times.

Hr	% of Peak	Avg Ontario Demand (MW)
1	75.47%	13265.24
2	73.20%	12866.76
3	72.01%	12657.11
4	71.82%	12623.87
5	73.25%	12874.23
6	77.21%	13570.89
7	83.31%	14643.22
8	88.27%	15515.62
9	90.17%	15849.33
10	90.96%	15987.43
11	91.71%	16119.22
12	92.45%	16249.04
13	92.58%	16273.37
14	92.51%	16260.37
15	92.80%	16310.68
16	94.69%	16643.63
17	98.03%	17230.55
18	100.00%	17576.71
19	99.69%	17522.97
20	98.02%	17228.52
21	95.25%	16741.53
22	90.38%	15885.76
23	84.31%	14819.50
24	78.95%	13876.85

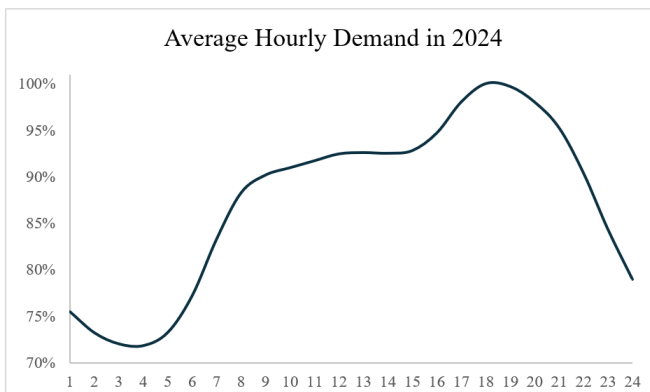

Figure 4: Data and Graph of Hourly Load Curve as a Percentage of Peak Load

1.3 Photovoltaic Farm (PV) Size and Reactive Power Curve Calculations

Various PV farm sizes to be used in the studies were determined and calculated based on data from the IESO [5] and IEEE Standard 2800-2022 [6]. See Table 2 for more design details. Three reactive power curves ("Q curves") were designed based on data from the IESO [7,8]. These curves dictate the ability of the PV farms in the simulations to lower the bus voltages of the Dx feeder and IEEE-14 bus systems to 0.97 per unit. Refer to Figure 5 for more details.

Table 2: Design and Calculation of Distribution (Dx) and Transmission (Tx) Connected PV Farms

Dx Inverter Size	Contract Capacity (MW)*	S base (MVA)**
Large	10	10/0.95 = 10.526
Small	5	5/0.95 = 5.263

Tx Inverter Size	Contract Capacity (MW)*	S base (MVA)**
Large	100	100/0.95 = 105.263
Medium	50	50/0.95 = 52.632
Small	10	10/0.95 = 10.526

*Contract capacity values determined from data available in [5]

**IEEE Standard 2800-2022 specifies that PV farm inverters must operate at a minimum power factor of 0.95 [6]. Therefore the MVA rating of the PV farm inverters were estimated by dividing contract capacity in MW by the power factor to get the rated apparent power in MVA.

	Sunny Day		Moderately Sunny Day		Cloudy Day	
Hour of the day	Active power in p.u. *	Corresponding Q (p.u.) available **	Active power in p.u. *	Corresponding Q (p.u.) available**	Active power in p.u. *	Corresponding Q (p.u.) available**
1	0.000	1.000	0.000	1.000	0.000	1.000
2	0.000	1.000	0.000	1.000	0.000	1.000
3	0.000	1.000	0.000	1.000	0.000	1.000
4	0.000	1.000	0.000	1.000	0.000	1.000
5	0.000	1.000	0.000	1.000	0.000	1.000
6	0.006	1.000	0.000	1.000	0.000	1.000
7	0.063	0.998	0.002	1.000	0.000	1.000
8	0.237	0.972	0.044	0.999	0.007	1.000
9	0.448	0.894	0.150	0.989	0.022	1.000
10	0.617	0.787	0.250	0.968	0.052	0.999
11	0.734	0.679	0.323	0.946	0.087	0.996
12	0.786	0.619	0.357	0.934	0.116	0.993
13	0.771	0.637	0.322	0.947	0.116	0.993
14	0.746	0.666	0.303	0.953	0.101	0.995
15	0.663	0.749	0.209	0.978	0.070	0.998
16	0.532	0.847	0.103	0.995	0.045	0.999
17	0.354	0.935	0.030	1.000	0.028	1.000
18	0.167	0.986	0.006	1.000	0.007	1.000
19	0.050	0.999	0.000	1.000	0.000	1.000
20	0.008	1.000	0.000	1.000	0.000	1.000
21	0.000	1.000	0.000	1.000	0.000	1.000
22	0.000	1.000	0.000	1.000	0.000	1.000
23	0.000	1.000	0.000	1.000	0.000	1.000
24	0.000	1.000	0.000	1.000	0.000	1.000

*Calculated by averaging MW values across various days in [7]and then dividing by the total MW capacity of transmission connected solar farms in Ontario to convert into per unit (478 MW) [8]:

$$Active\ power\ in\ p.u. = \frac{(Day\ 1\ MW + Day\ 2\ MW + Day\ 3\ MW + Day\ 4\ MW)}{478\ MW}$$

Distribution connected solar farms do not have publically disclosed data on their hourly generation. IESO only reports on tranmission connected ones.

**Calculated by using the following relationship between apparent power (S), active power (P) and reactive power (Q):

$$Corresponding\ Q\ (p.u.)\ available = \sqrt{S^2 - P^2} = \sqrt{1^2 - P\ (p.u.)^2}$$

Note that S is equal to 1 per unit in this case.

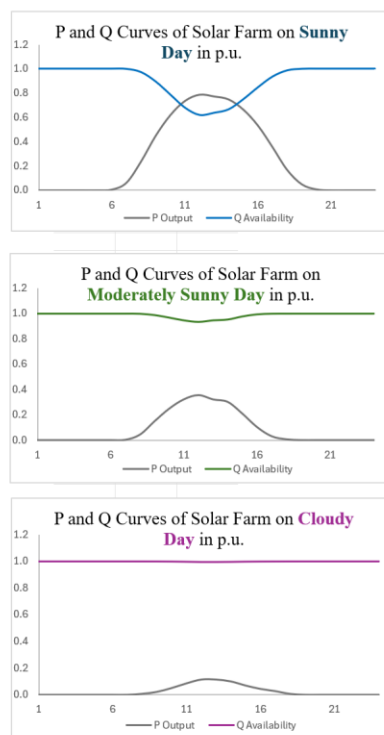

Figure 5: Q Curve Calculations and Graphs

1.4 Percent Reduction Equation

To analyze the effectiveness of CVR on the two networks, a "before and after" analysis was conducted on every different condition tested. The load reduction percentage is defined as the following: $(MW_{after\ CVR} - MW_{before\ CVR})/\ MW_{before\ CVR}$ where "MW" refers to the load in the unit of megawatts of Bus 5 in the Dx feeder and buses 4, 9 and 14 in the IEEE 14-bus system.

1.5 AI Model Design

The AI component was developed as a surrogate CVR recommendation model using PSSE simulation outputs to predict next-day feeder performance under different operating scenarios. A baseline-plus-delta formulation was used, where the model predicts the no-CVR state and then separately predicts the CVR-induced change. This structure preserves physically consistent behaviour by ensuring that CVR reduces both load and voltage. Refer to Figure 6 throughout for a design summary of the AI Model.

Component	Description
Model purpose	Surrogate CVR recommendation model using PSSE simulation outputs to predict next-day feeder performance under different operating scenarios.
Predicted quantities	Baseline feeder load, baseline load-bus voltage, MW reduction due to CVR, and voltage drop due to CVR.
Final formulation	Baseline-plus-delta approach: predict the no-CVR state, then separately predict the CVR-induced change.
Key advantage	Enforces physically consistent behaviour by ensuring CVR reduces both load and voltage and avoids impossible outputs.
Reconstruction equations	P_cvr = P_base - ΔP ; V_cvr = V_base - ΔV

Figure 6: AI Model Design Summary

Three model families were evaluated for the final design: Random Forest, Extra Trees, and Histogram Gradient Boosting. Models were evaluated using grouped cross-validation and a grouped held-out test set, where all 24 hours of a given operating scenario were kept together in either training or testing to prevent data leakage between similar cases. The evaluation metrics were Mean Absolute Error (MAE) and the coefficient of determination (R^2). Random Forest and Extra Trees both outperformed Histogram Gradient Boosting overall on these metrics and therefore were selected for a weighted ensemble blend of 60% Extra Trees and 40% Random

Forest as the final model (see Figures 7 and 8 for details on the evaluation results of the final model chosen).

Criterion	Requirement	Reported outcome
Grouped test accuracy	$R^2 \geq 0.95$ for all four visible targets	Satisfied
Grouped cross-validation accuracy	$R^2 \geq 0.90$ for all four visible targets	Satisfied
Overfitting check	Train R^2 − Test $R^2 < 0.05$ for all targets	Satisfied; all gaps below 0.002
MW reduction MAE check	Cross-validation MAE for CVR MW reduction < 0.01 MW	Satisfied

Figure 7: Validation Criteria and Outcomes of Final AI Model

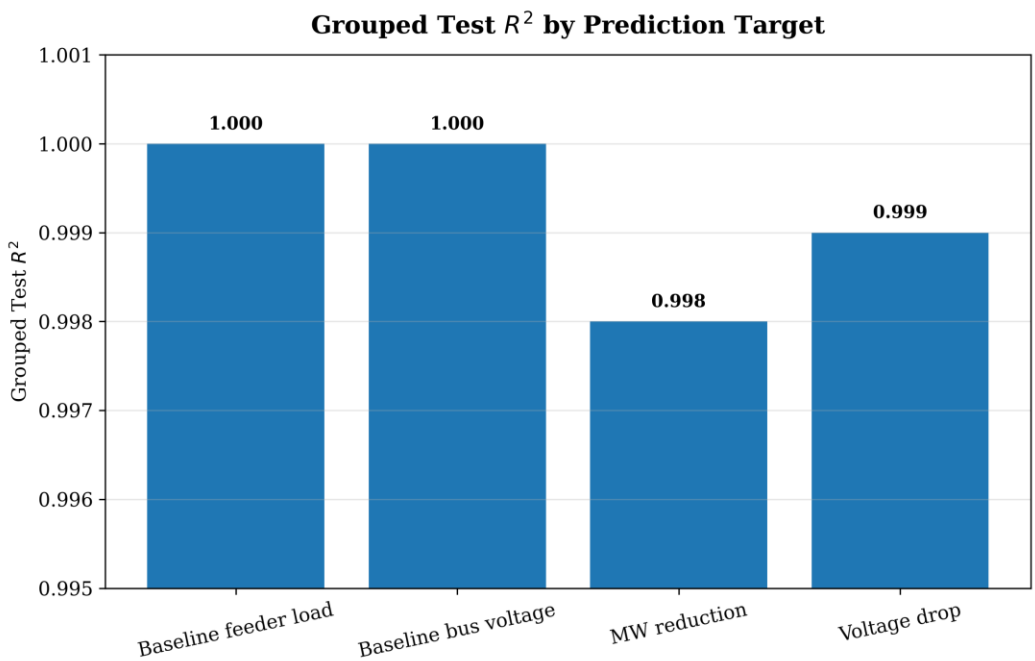

Figure 8: Test R^2 by Prediction Target

In this project, "effective" means more than a high R^2 value. The model was considered effective because it accurately predicted unseen grouped scenario cases, preserved monotonic CVR behaviour (with-CVR load never exceeds baseline), respected ANSI C84.1 voltage limits [1], and produced outputs suitable for ranking and selecting the best next-day operating scenario.

2. Prototype

2.1 Hardware Prototype: Cost and Construction

Our hardware (Figure 9) simulates the effect of an inductor acting as a solar farm on a small-scale system. Tx Line 1 represents a long transmission line, including resistance and inductance that would be associated with a standard line. Tx Line 2 represents a shorter transmission line, with a smaller resistance, and negligible inductance. The load of the system is represented as a pure resistance, as CVR works best at reducing resistive loads. The simulated PV farm is represented by an inductor, and a resistor in series to throttle the current allowed to pass through the inductor. The dual throw switch allows the simulated solar farm to be connected at either the midline bus, or the load bus. Each of the measure points is a critical point in the system. Point 1 is the incoming voltage. Point 2 is the midline voltage and point 3 is the bus voltage. When in use, this prototype will show how voltage in larger scale systems can be reduced.

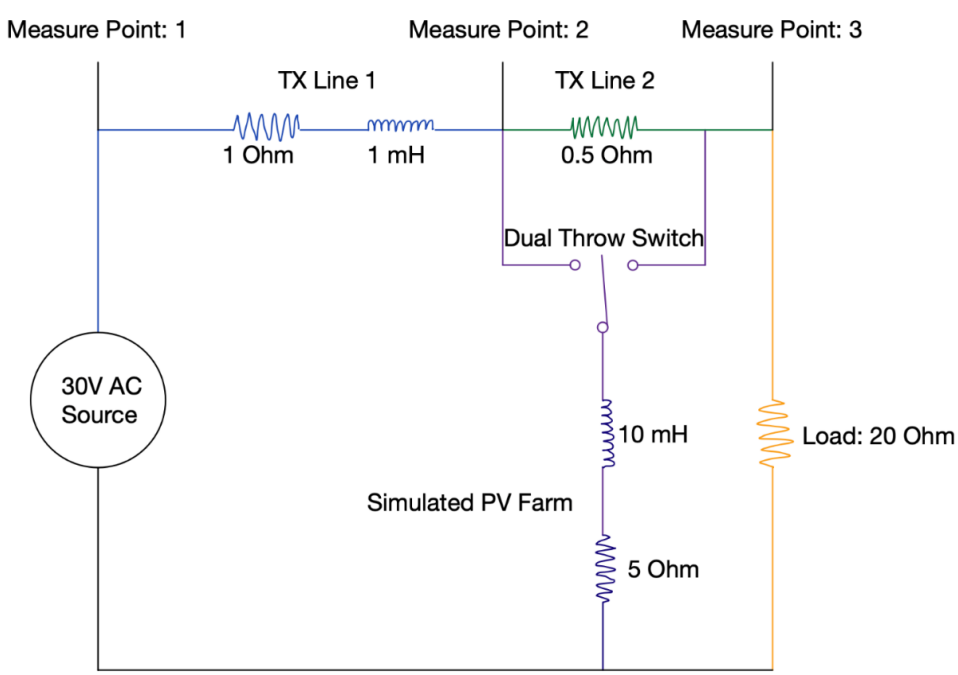

Figure 9: Schematic of Hardware Circuit

Refer to Table 3 below for the list of components to construct the circuit and their costs.

Table 3: Components and Costs List of Hardware

Component	Value	Number Required	Cost ($)
TX Line 1 Resistor	1 Ohm, 100W	1	7.5
TX Line 1 Inductor	1 mH, 5A	1	2.29
TX Line 2 Resistor	0.5 Ohm, 100W	1	5.15
PV Inductor	10 mH, 5A	1	65.21
PV Resistor	5 Ohm, 100W	1	6.65
Load Resistor	20 Ohm. 100W	1	4.75
Dual throw Switch	125 VAC, 20A	1	5.2
Measuring Device	NA	3	NA
Power Supply	30VAC, 150VA	1	60

2.2 Software Prototype: Python Automation Flow Charts

See Figures 10 and 11 for details on the flow of the Python scripts written to automate and run the load flow studies. Note that they are similar with only slight differences.

2.2.1 Dx Feeder Python Automation Flow Chart

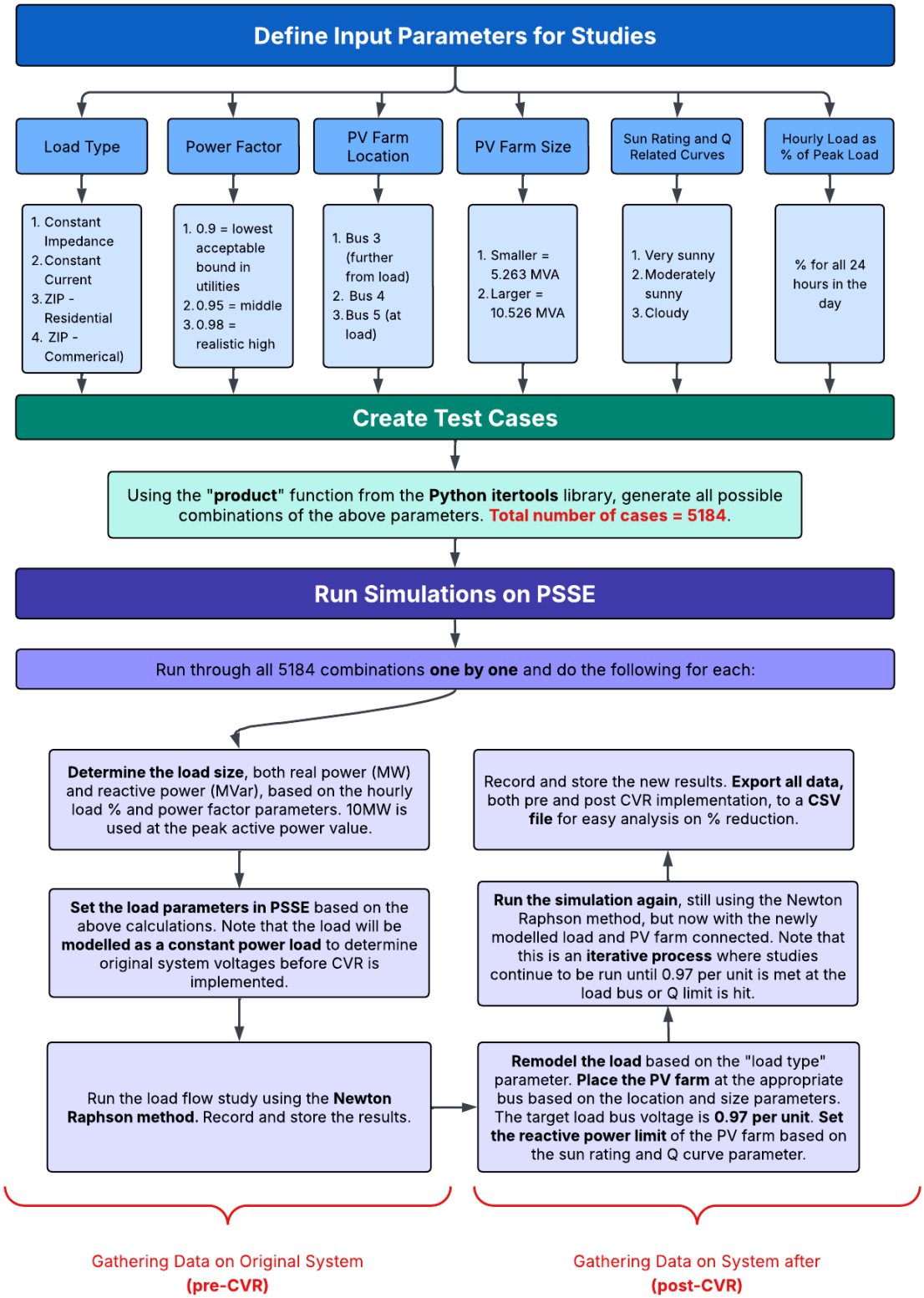

Figure 10: Program Flow for Dx Feeder Python Script

2.2.2 IEEE 14-Bus System Automation Flow Chart

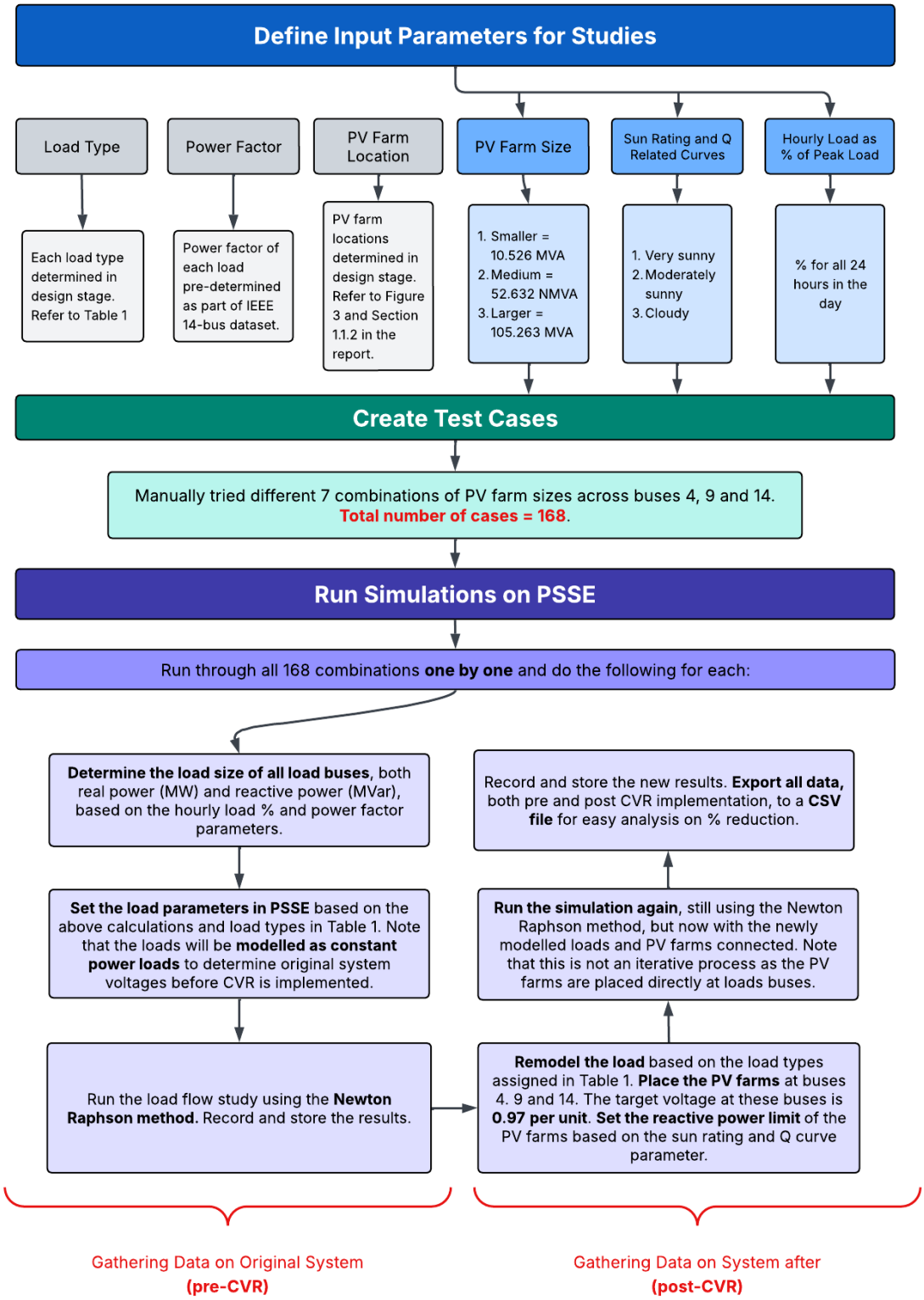

Figure 11: Program Flow for IEEE 14-bus System Python Script

2.3 AI Prototype: Model Details

The AI prototype is a software-based decision-support tool built on PSSE study results. It learns the relationship between operating conditions and CVR response, then recommends the best next-day operating scenario without requiring a new simulation run. See Figure 12 below for the workflow of the AI model:

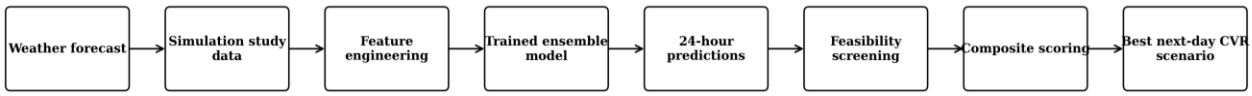

Figure 12: AI Model Workflow

The training dataset was built from various sources covering different load types: Constant-Z load, Constant-I load, and ZIP load study outputs. To make the recommendation relevant to the following day, the model integrates hourly weather forecast data retrieved from the Open-Meteo API (temperature, cloud cover, rain, and snowfall) and matches the forecast to the most similar historical operating scenario. The model outputs a ranked list of candidate scenarios evaluated for voltage feasibility [1], MW reduction, and economic value alongside summary graphs of these results. The top-ranked scenario provides a 24-hour profile of predicted feeder load with and without CVR, bus voltage, energy savings, and peak demand value for use in the Streamlit dashboard.

3. Validation/Testing Strategy or Protocols

3.1 Validation of Studies

The objective of the validation strategy is to verify that the proposed CVR design using PV inverter reactive power control can reduce feeder demand while still operating safe conditions. This project evaluates CVR effectiveness by comparing system demand before and after voltage reduction. The primary performance metric used for validation is percent load reduction, which measures the effectiveness of CVR in reducing demand and is calculated using the equation in Section 1.4. As seen in Figures 10 and 11, load flow studies were performed in PSSE using a Python script to automate simulations across a wide range of operating conditions. The script varied parameters such as hourly demand, solar irradiance, PV size, reactive power limits, and PV bus location. Testing combinations of these conditions allowed the design to be evaluated under representative feeder operating scenarios rather than a single operating point.

The primary validation requirement defined in the design specifications is that the CVR strategy must achieve at least a 2% reduction in feeder demand under some circumstances while maintaining acceptable voltage levels. Simulation results demonstrated that the system consistently achieves reductions on the order of 2-3%. Figure 13 shows a summary graph of the validation analysis done for the constant impedance load model, with an average reduction of 2.36%, exceeding the 2% reduction requirement defined in the design specifications.

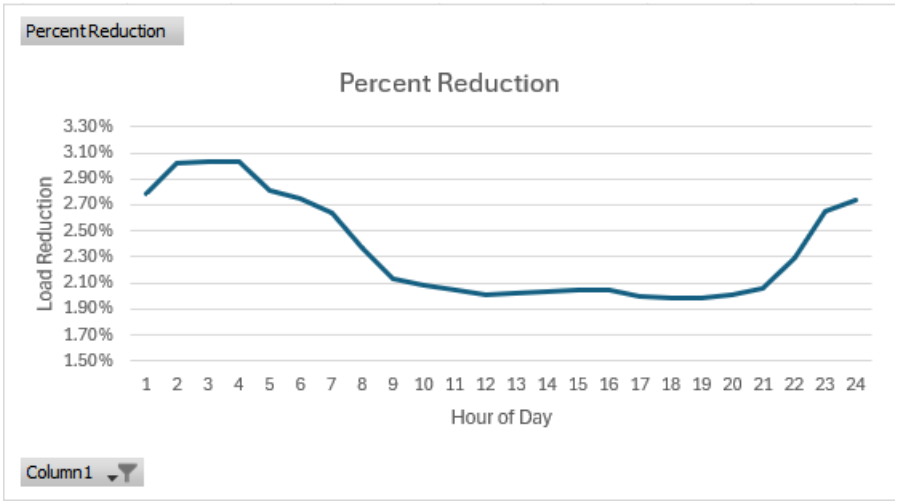

Figure 13: Hourly CVR % MW Reduction across a 24-hour demand cycle for Constant Impedance Loads

In addition to demand reduction, relevant simulation parameters were set to ensure that the bus voltages remain within the acceptable operating range of 0.95 - 1.05 per unit, ensuring that voltage reductions do not cause undervoltage conditions. Our testing strategy provided a systematic approach for validating the proposed design and demonstrated that the CVR strategy meets the project objectives by achieving the required demand reduction while maintaining system operating constraints.

3.2 Validation of AI Model

The AI validation strategy was designed to confirm that the model generalizes unseen operating scenarios and that its predictions satisfy the physical and engineering constraints required for CVR operation. Each operating scenario produces a full 24-hour load profile, grouped train/test splitting and grouped cross-validation were used throughout. All 24 hours of a scenario were assigned entirely to either training or testing to prevent leakage and ensure true generalization.

The statistical validation criteria and engineering feasibility conditions are summarized in Figures 13 and 14.

A scenario was considered feasible only if all hourly constraints were satisfied, and the daily aggregated load reduction was at least 2.0%, as required by the design specification. Together, these checks confirmed that the AI model produces accurate, physically valid, and voltage-safe predictions for next-day CVR decision-making. The preliminary AI results shown in Figures 15 and 16 demonstrate that the surrogate model accurately reproduced PSSE simulation outputs for unseen operating scenarios while maintaining expected CVR behaviour, validating the model.

Model family	Observed performance	Used in final model?
Random Forest	Strongest on baseline MW prediction; selected for ensemble	Yes
Extra Trees	Lower MAE on delta targets; selected for ensemble	Yes
Histogram Gradient Boosting	Underperformed RF and ET on CVR-response targets	No

Figure 14: Benchmark Model Families and Final Ensemble Selection

Check	Condition	Purpose
Non-negativity	$\Delta P(h) \geq 0$ for all $h \in \{1, \ldots, 24\}$	Ensures predicted MW reduction is physically valid
Monotonicity	$P_cvr(h) \leq P_base(h)$ for all $h \in \{1, \ldots, 24\}$	Preserves expected CVR behaviour
Voltage bounds	$V_cvr(h) \in [0.95, 1.05]$ pu for all $h \in \{1, \ldots, 24\}$	Satisfies ANSI C84.1 operating limits
Daily design requirement	Daily aggregated load reduction $\geq 2.0\%$	Meets project design specification

Figure 15: Engineering Feasibility Conditions

— 24-Hour Hourly Detail —

hour	pred_load_mw_no_cvr	pred_load_mw_with_cvr	pred_delta_mw	pred_reduction_pct	pred_load_bus_v_no_cvr_pu	pred_load_bus_v_with_cvr_pu	pred_energy_savings_$	pred_total_value_$
1	7.7247	7.0337	0.6910	8.9455	1.0168	0.9708	51.8262	51.8262
2	7.5554	6.8704	0.6851	9.0670	1.0176	0.9709	51.3790	51.3790
3	7.4659	6.7826	0.6833	9.1519	1.0176	0.9709	51.2451	51.2451
4	7.4622	6.7784	0.6838	9.1635	1.0176	0.9709	51.2851	51.2851
5	7.5946	6.9062	0.6884	9.0646	1.0176	0.9709	51.6313	51.6313
6	8.0411	7.3400	0.7010	8.7183	1.0154	0.9708	52.5786	52.5786
7	8.6304	7.9250	0.7054	8.1733	1.0116	0.9699	74.0657	74.0657
8	9.1039	8.3979	0.7060	7.7545	1.0100	0.9700	74.1254	116.4827
9	9.1833	8.4777	0.7056	7.6832	1.0098	0.9701	74.0854	74.0854
10	9.2360	8.5305	0.7055	7.6389	1.0091	0.9701	74.0805	74.0805
11	9.2640	8.5585	0.7055	7.6158	1.0091	0.9701	74.0804	74.0804
12	9.2956	8.5903	0.7053	7.5875	1.0090	0.9701	74.0569	74.0569
13	9.3241	8.6192	0.7050	7.5607	1.0090	0.9701	74.0218	74.0218
14	9.3488	8.6439	0.7048	7.5392	1.0090	0.9701	74.0067	74.0067
15	9.3963	8.6923	0.7040	7.4922	1.0087	0.9700	73.9189	73.9189
16	9.5758	8.8738	0.7020	7.3309	1.0078	0.9703	73.7090	73.7090
17	9.8638	9.1682	0.6956	7.0516	1.0060	0.9700	97.3776	97.3776
18	9.9876	9.2937	0.6940	6.9481	1.0053	0.9700	97.1531	97.1531
19	9.9570	9.2625	0.6945	6.9748	1.0060	0.9700	97.2266	97.2266
20	9.8459	9.1480	0.6980	7.0887	1.0061	0.9700	97.7135	97.7135
21	9.5903	8.8876	0.7027	7.3267	1.0073	0.9699	73.7785	73.7785
22	9.1148	8.4098	0.7050	7.7346	1.0101	0.9698	74.0240	74.0240
23	8.5165	7.8147	0.7018	8.2408	1.0133	0.9704	52.6371	52.6371
24	8.0652	7.3702	0.6950	8.6177	1.0150	0.9707	52.1273	52.1273

— Top 5 Candidate Scenarios —

dataset_type	sun_rating	temperature_c	pv_bus	pf	overall_reduction_pct	total_economic_value_$	min_v_with_cvr_pu	feasible	selection_score
ConstantZ	cloudy	15.0	5	0.98	7.8671	1734.4911	0.9698	True	0.8234
ConstantZ	cloudy	15.0	5	0.98	7.6385	1696.4073	0.9699	True	0.8052
ConstantZ	cloudy	15.0	4	0.98	6.6212	1458.6051	0.9717	True	0.7179
ConstantZ	cloudy	15.0	4	0.98	5.2551	1180.4718	0.9785	True	0.6463
ConstantZ	cloudy	15.0	5	0.95	6.1616	1344.9280	0.9682	True	0.6424

Figure 16: Detailed Output Results of AI Model Validation

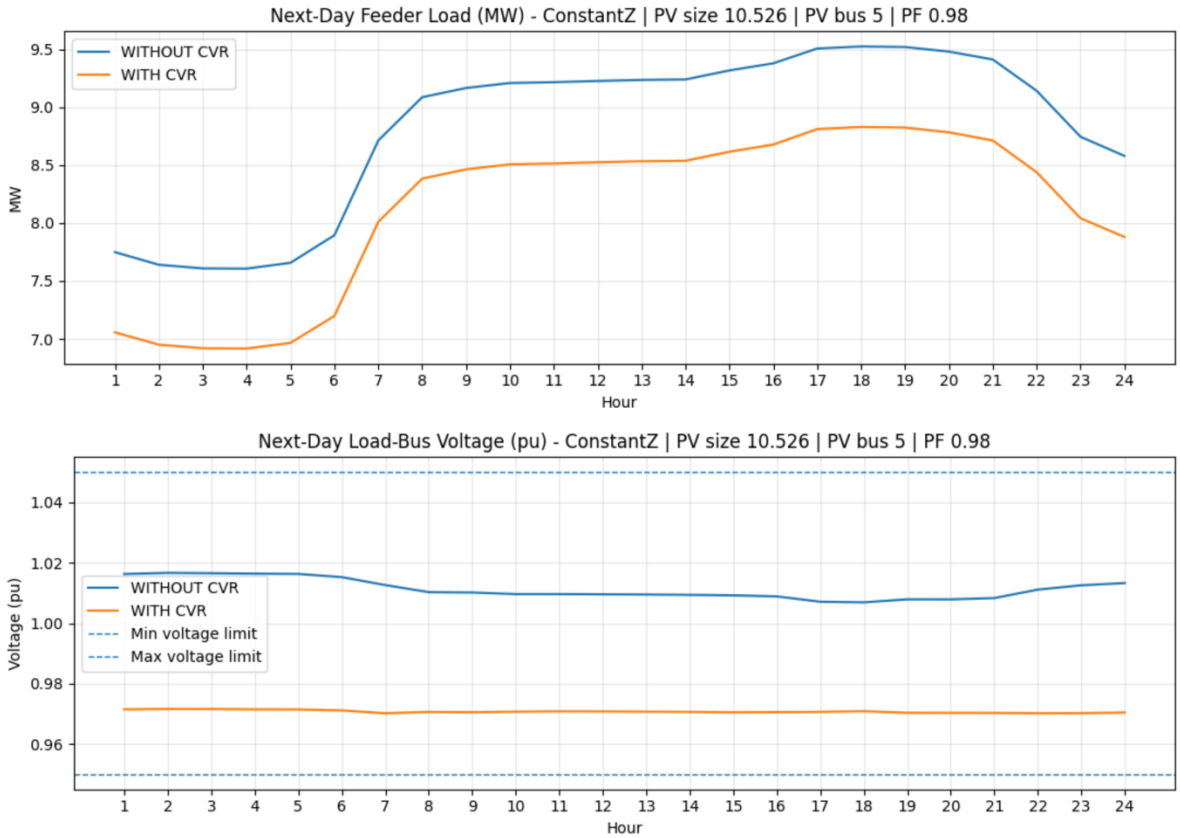

Figure 17: Load and Voltage Results of AI Model Validation

3.3 Hardware Testing

To verify that the prototype does in fact work, it will first need to be constructed. All online simulations show that the prototype will function and reduce the voltage by a notable amount. When the prototype is constructed, it will need to show significant reductions at the midline and load buses. The voltage will then be dropped further when the inductor is introduced to the circuit. The amount that the voltage drops will be dependent on which bus the inductor is connected to. To summarize, the physical prototype, if it behaves like the simulated circuit, will work and demonstrate what it is supposed to. If not, then the prototype will have to be re-evaluated.

4. Preliminary Test Results

4.1 Preliminary Study Results

Preliminary results were obtained from the automated PSSE load flow simulations described in the validation strategy. These studies evaluated system performance under various operating conditions, including changes in load level, PV size, solar irradiance, reactive power limits, and

PV bus locations. The effectiveness of the proposed CVR strategy was evaluated using the percentage reduction in feeder load, which directly measures the impact of voltage reduction on system demand.

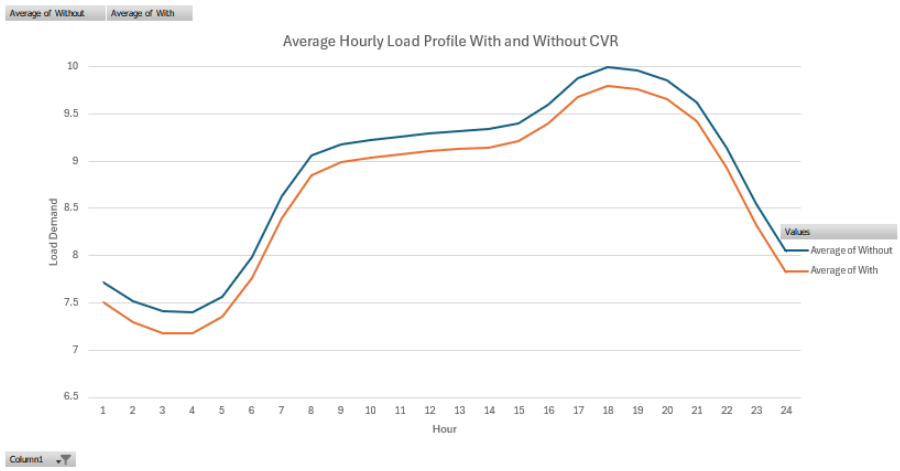

Figure 18: Average Hourly Load Profile of Simulations With and Without CVR

Figure 18 shows the average hourly load profile with and without CVR across the 24-hour cycle. The results indicate that applying CVR consistently lowers feeder demand throughout the day while preserving the overall daily load shape. This confirms that the proposed control strategy is effective in reducing system demand rather than altering the natural load pattern. The difference between the two curves represents the load savings achieved through voltage reduction.

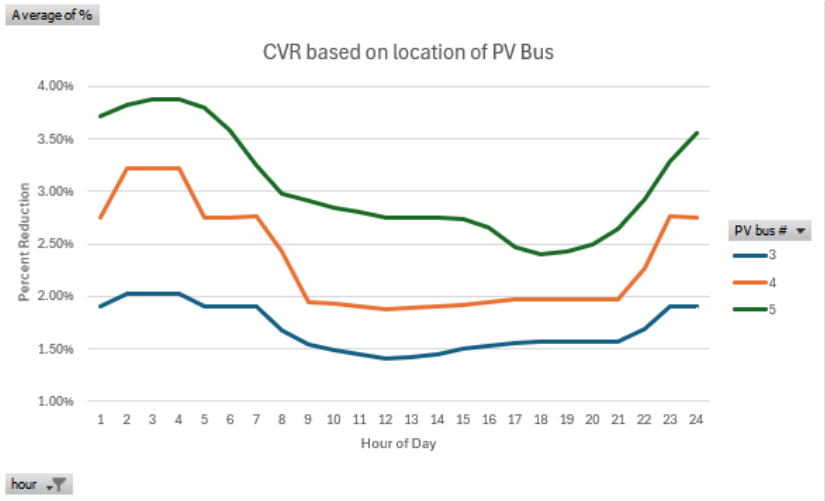

Figure 19: Hourly % Reduction in Load Based on PV Bus Location

Additional studies were conducted to determine how system parameters influence CVR performance. Figure 19 shows the impact of PV bus location on CVR performance. The results

indicate that PV placement significantly affects the magnitude of reduction achieved. PV connected further downstream provides stronger local voltage support at buses that are more vulnerable to voltage drop, allowing the CVR strategy to reduce demand more effectively while remaining within acceptable voltage limits. In contrast PV connected closer to the substation produces lower percentage reduction because its influence on downstream voltage support is smaller.

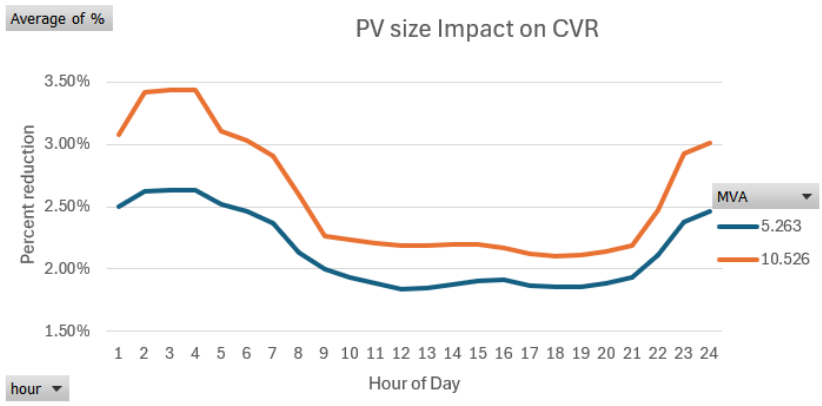

Figure 20: Hourly % Reduction in Load Based on PV Inverter Size

Figure 20 shows the impact of PV size on CVR effectiveness. The larger PV size consistently produces greater percentage reduction than the smaller PV configuration, indicating that increased PV penetration provides additional voltage support and creates more flexibility for CVR operation. This suggests that feeder voltage margin plays an important role in determining achievable demand reduction.

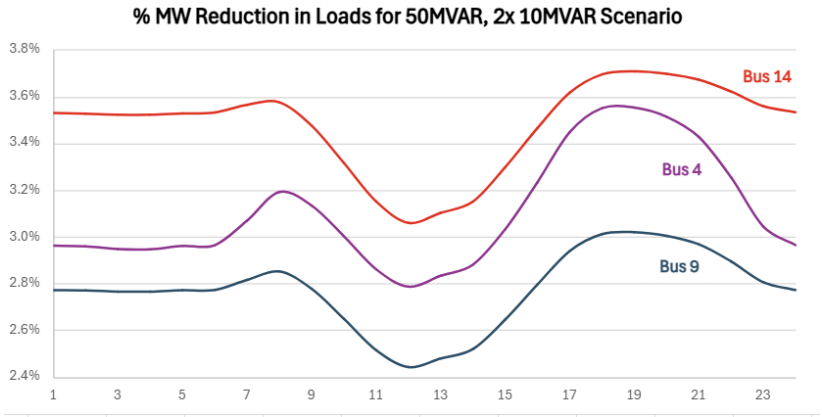

Figure 21: Hourly % Reduction on IEEE 14-Bus System with Two 50MVAR PV Farms and One 10MVAR PV Farm

To further evaluate the robustness of the proposed concept, CVR performance was also tested on the IEEE 14-bus system (Figure 5). The results show that load reduction varies depending on the bus where reactive power support is applied, with Bus 14 achieving the highest MW reduction, followed by Bus 4 and Bus 9. This demonstrates that bus location significantly affects CVR performance. The larger reduction at Bus 14 suggests that electrically weaker or downstream buses benefit more from voltage support. These results strengthen the validation by showing that the previously observed trends extend beyond a single feeder configuration. Overall, the preliminary studies confirm that the proposed CVR strategy consistently achieves the targeted demand reduction while maintaining acceptable operating conditions. The results also show that CVR effectiveness is influenced by PV size, PV bus location, and network structure, providing useful insight into how distributed generation and reactive power control can be coordinated to improve distribution system performance

4.2 Preliminary AI Model Results

The preliminary AI results showed that the surrogate model accurately reproduced PSSE simulation outputs for unseen operating scenarios while maintaining expected CVR behaviour. Using the next-day weather forecast, the AI scheduler matched conditions to the selected scenario seen in Figure 22 and predicted an average reduction of 7.87% and an estimated daily cost saving of $1,713.44 (Figure 23). These results show that the AI component can identify a physically valid and economically beneficial next-day CVR operating plan from specified inputs.

Metric	Value
Matched scenario type	Constant-Z scenario
PV bus	5
Power factor	0.98
PV size	10.526 MVA
Average baseline load	8.88 MW
Daily load reduction	7.87%
Energy savings	16.77 MWh
Minimum with-CVR bus voltage	0.9698 pu
Estimated daily economic value	$1,713.44

Figure 22: Selected Next-Day AI Scenario Summary

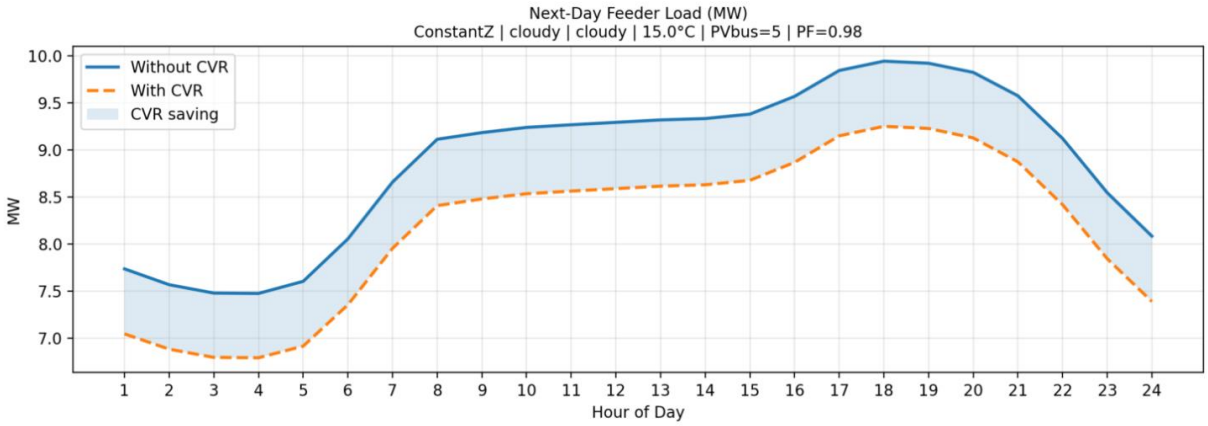

Figure 23: Next-Day Feeder Load Graph for Selected Scenario

5. Conclusion

Our capstone project is near completion. All intended load flow studies were completed, and their results validated the method. The surrogate AI model was also developed to predict CVR performance for next-day operating conditions. Two major changes were made during the project. First, the simulation platform was changed from PowerWorld to PSSE, which allowed a

larger number of automated simulation cases to be evaluated. Second, the hardware prototype was redesigned from a scaled 120 V system to a simplified 30 V demonstration circuit to meet safety and cost constraints while still illustrating the principles of CVR. The next steps of the project include constructing and testing the hardware prototype, consolidating the simulation results to identify where the greatest demand reductions occur, and developing a dashboard to clearly present the system behaviour and AI recommendations.

6. References

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